

● HARD

● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

07

10 questions · ~15 min

Q1

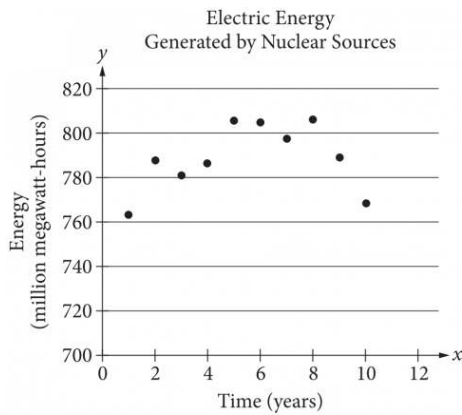
PROBLEM-SOLVING AND DATA ANALYSIS

The area of a rectangular region is increasing at a rate of 250 square feet per hour. Which of the following is closest to this rate in square meters per minute?

Use 1 meter = 3.28 feet.

- A. 0.39
- B. 1.27
- C. 13.67
- D. 23.24

The scatterplot below shows the amount of electric energy generated, in millions of megawatt-hours, by nuclear sources over a 10-year period.



Of the following equations, which best models the data in the scatterplot?

- A. $y = 1.674x^2 + 19.76x - 745.73$
- B. $y = -1.674x^2 - 19.76x - 745.73$
- C. $y = 1.674x^2 + 19.76x + 745.73$
- D. $y = -1.674x^2 + 19.76x + 745.73$

Q3

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

The regular price of a shirt at a store is \$ 11.70. The sale price of the shirt is 80% less than the regular price, and the sale price is 30% greater than the store's cost for the shirt. What was the store's cost, in dollars, for the shirt? (Disregard the \$ sign when entering your answer. For example, if your answer is \$ 4.97, enter 4.97)

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Each of the following frequency tables represents a data set. Which data set has the greatest mean?

A.

Value	Frequency
70	4
80	5
90	6
100	7

B.

Value	Frequency
70	6
80	6
90	6
100	6

C.

Value	Frequency
70	7
80	6
90	6
100	7

D.

Value	Frequency
70	8
80	5

90	5
100	8

Q5

PROBLEM-SOLVING AND DATA ANALYSIS

The speed of a vehicle is increasing at a rate of 7.3 meters per second squared. What is this rate, in **miles per minute squared**, rounded to the nearest tenth?

Use 1 mile = 1,609 meters.

- A. 0.3
- B. 16.3
- C. 195.8
- D. 220.4

Q6

PROBLEM-SOLVING AND DATA ANALYSIS

37% of the items in a box are green. Of those, 37% are also rectangular. Of the green rectangular items, 42% are also metal. Which of the following is closest to the percentage of the items in the box that are not rectangular green metal items?

- A. 1.16%
- B. 57.50%
- C. 94.25%
- D. 98.84%

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

A scientist measured the lengths of 240 gray seals from Muskeget Island and 120 gray seals from Sable Island. The scientist determined that the mean length of the 240 gray seals from Muskeget Island was 88 inches and the mean length of the 120 gray seals from Sable Island was 94 inches. What was the mean length of all 360 gray seals the scientist measured for this study?

- A. 89
- B. 90
- C. 91
- D. 92

A sample of oak has a density of 807 kilograms per cubic meter. The sample is in the shape of a cube, where each edge has a length of 0.90 meters. To the nearest whole number, what is the mass, in kilograms, of this sample?

- A. 588
- B. 726
- C. 897
- D. 1,107

Q9

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

The number a is 70% less than the positive number b . The number c is 60% greater than a . The number c is how many times b ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

	Masses (kilograms)					
Andrew	2.4	2.5	3.6	3.1	2.5	2.7
Maria	x	3.1	2.7	2.9	3.3	2.8

Andrew and Maria each collected six rocks, and the masses of the rocks are shown in the table above. The mean of the masses of the rocks Maria collected is 0.1 kilogram greater than the mean of the masses of the rocks Andrew collected. What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	/	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.